Assignment 11

1.Draw the OSI model on paper or using any drawing tool, label all 7 layers, and write one real-world example protocol (like HTTP, TCP, IP, etc.) next to each layer.

Ans-

Layer	Name	Example	Main Work
7	Application	HTTP	User/application communication
6	Presentation	TLS/SSL	Encryption & data formatting
5	Session	NetBIOS	Establish/manage sessions
4	Transport	TCP/UDP	Reliable delivery / segmentation
3	Network	IP	Routing between networks
2	Data Link	Ethernet	Frames & MAC addresses
1	Physical	Wi-Fi	Sends bits as signals

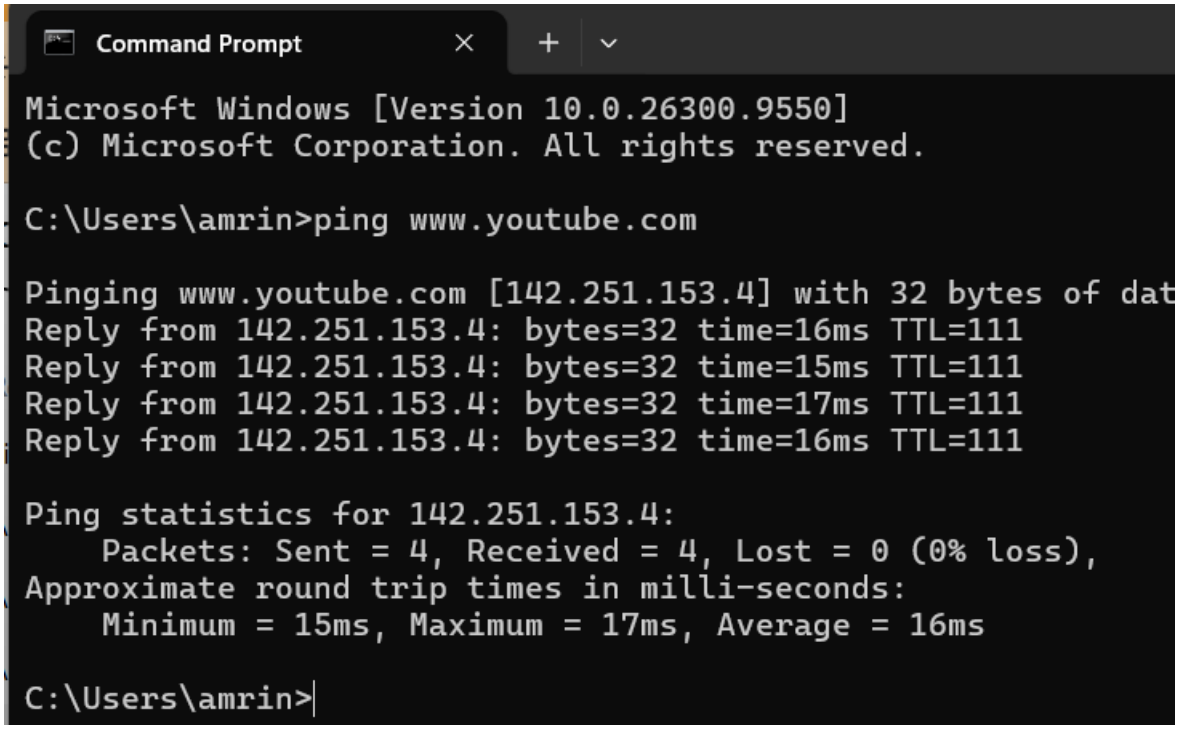
2.Create a table listing 5 popular apps you use (such as WhatsApp, Instagram, Zomato, Flipkart, Paytm) and for each, identify two network protocols that are likely used when you interact with the app (for example, HTTPS, TCP, UDP, DNS, etc.).

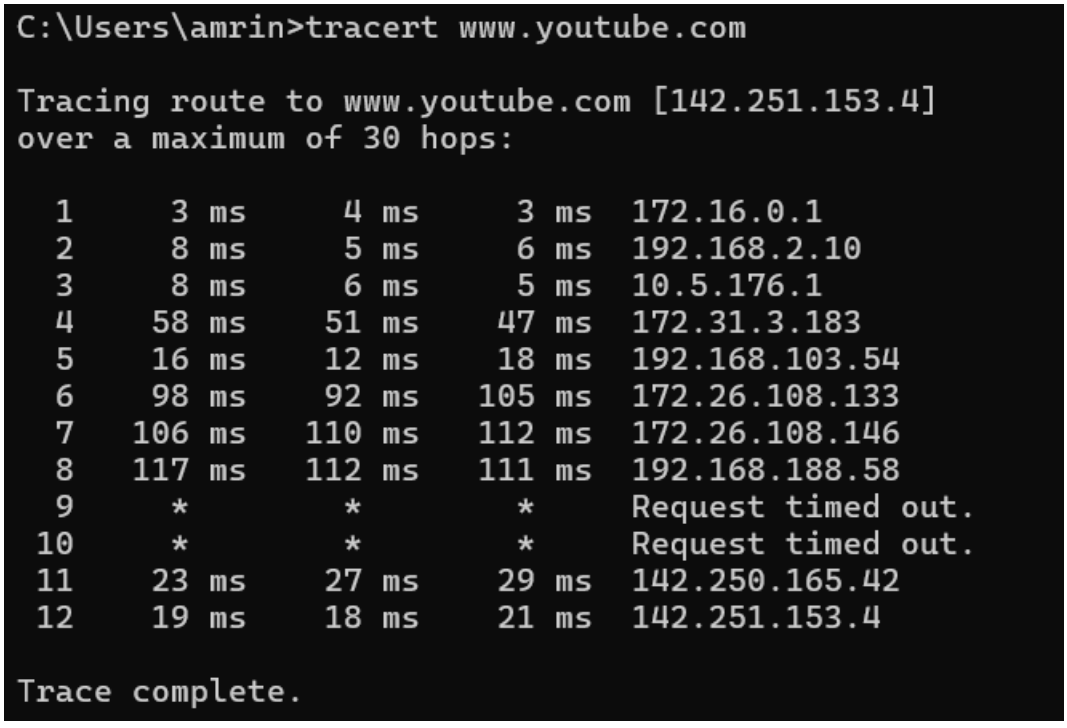
Ans-

No.	App	Protocol 1	Protocol 2	Use
1	WhatsApp	HTTPS	TCP	Secure communication & reliable data
2	Instagram	HTTPS	DNS	Web/API communication & domain lookup
3	Zomato	HTTPS	TCP	Secure app communication
4	Flipkart	HTTPS	DNS	Secure communication & domain resolution
5	Paytm	HTTPS	TLS	Secure communication & encryption

3.Use the command prompt or terminal to run 'ping' and 'tracert' (Windows) or 'traceroute' (Linux/Mac) to www.youtube.com, then explain which OSI layers are involved in these commands and how.

Ans-





**4.Write a short scenario describing how a message sent from your device to a friend's device on WhatsApp travels through each OSI layer, mentioning what happens at each step.

Hint: Start from Application layer (your message in WhatsApp) down to Physical layer (actual data sent over WiFi/mobile network), then back up to Application layer on your friend's phone.**

Ans-

Step 1 — Application Layer (Layer 7)

You open WhatsApp and type:

Hello

WhatsApp prepares the message for transmission.

Layer: Application
Example: WhatsApp application protocols / HTTPS-related communication

Step 2 — Presentation Layer (Layer 6)

The data is processed into a suitable format and protected using encryption mechanisms.

WhatsApp uses end-to-end encryption for messages.

Layer: Presentation
Work: Encryption/data representation

Step 3 — Session Layer (Layer 5)

The communication session between the application and its services is maintained.

Layer: Session
Work: Establishing and managing communication sessions

Step 4 — Transport Layer (Layer 4)

The data is handled by transport protocols such as TCP or UDP, depending on the communication function.

Layer: Transport
Work: Data delivery, segmentation, ports

Step 5 — Network Layer (Layer 3)

IP addressing is used to move packets between networks.

Your Phone IP

↓

Router/ISP

↓

Internet

↓

Friend's Network

Layer: Network
Example: IP

Step 6 — Data Link Layer (Layer 2)

Your phone uses Wi-Fi or mobile-network link technology to transmit data over the local connection.

MAC/link-layer addressing and frames are involved where applicable.

Layer: Data Link
Example: Wi-Fi/Ethernet

Step 7 — Physical Layer (Layer 1)

The bits are transmitted as actual signals:

- Wi-Fi → radio waves
- Ethernet → electrical/optical signals
- Mobile network → radio signals

Layer: Physical